Q1. JEE Main 2026 (21 January Shift 2)

For a triangle ABC, let $\vec{p}=\overrightarrow{BC}, \vec{q}=\overrightarrow{CA}$ and $\vec{r}=\overrightarrow{BA}$. If $|\vec{p}|=2\sqrt{3}, |\vec{q}|=2$ and $\cos\theta=\frac{1}{\sqrt{3}}$, where θ is the angle between $\vec{p}$ and $\vec{q}$, then $|\vec{p}\times(\vec{q}-3\vec{r})|^2+3|\vec{r}|^2$ is equal to :

(1) 410 (2) 340 (3) 200 (4) 220

Q2. JEE Main 2026 (23 January Shift 1)

Let $\vec{a}=-\hat{i}+\hat{j}+2\hat{k}, \vec{b}=\hat{i}-\hat{j}-3\hat{k}, \vec{c}=\vec{a}\times\vec{b}$ and $\vec{d}=\vec{c}\times\vec{a}$. Then $(\vec{a}-\vec{b})\cdot\vec{d}$ is equal to :

(1) -4 (2) 4 (3) 2 (4) -2

Q3. JEE Main 2026 (28 January Shift 2)

Let P be a point in the plane of the vectors $\overrightarrow{AB}=3\hat{i}+\hat{j}-\hat{k}$ and $\overrightarrow{AC}=\hat{i}-\hat{j}+3\hat{k}$ such that P is equidistant from the lines AB and AC. If $|\overrightarrow{AP}|=\frac{\sqrt{5}}{2}$, then the area of the triangle ABP is:

(1) 2 (2) $\frac{\sqrt{30}}{4}$ (3) $\frac{3}{2}$ (4) $\frac{\sqrt{26}}{4}$

Q4. JEE Main 2026 (24 January Shift 2)

Let $\vec{a}=2\hat{i}-5\hat{j}+5\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+3\hat{k}$. If $\vec{c}$ is a vector such that $2(\vec{a}\times\vec{c})+3(\vec{b}\times\vec{c})=\vec{0}$ and $(\vec{a}-\vec{b})\cdot\vec{c}=-97$, then $|\vec{c}\times\hat{k}|^2$ is equal to

(1) 233 (2) 218 (3) 193 (4) 205

Q5. JEE Main 2026 (24 January Shift 2)

Let $\vec{a}=2\hat{i}-\hat{j}-\hat{k}, \vec{b}=\hat{i}+3\hat{j}-\hat{k}$ and $\vec{c}=2\hat{i}+\hat{j}+3\hat{k}$. Let $\vec{v}$ be the vector in the plane of the vectors $\vec{a}$ and $\vec{b}$, such that the length of its projection on the vector $\vec{c}$ is $\frac{1}{\sqrt{14}}$. Then $|\vec{v}|$ is equal to

(1) 13 (2) $\frac{\sqrt{35}}{2}$ (3) $\frac{\sqrt{21}}{2}$ (4) 7

Q6. JEE Main 2026 (22 January Shift 2)

Let $\vec{a}=2\hat{i}-\hat{j}+\hat{k}$ and $\vec{b}=\lambda\hat{j}+2\hat{k}, \lambda\in\boldsymbol{Z}$ be two vectors. Let $\vec{c}=\vec{a}\times\vec{b}$ and $\vec{d}$ be a vector of magnitude 2 in yz-plane. If $|\vec{c}|=\sqrt{53}$, then the maximum possible value of $(\vec{c}\cdot\vec{d})^2$ is equal to :

(1) 26 (2) 52 (3) 208 (4) 104

Q7. JEE Main 2026 (22 January Shift 2)

Let a vector $\vec{a}=\sqrt{2}\hat{i}-\hat{j}+\lambda\hat{k}, \lambda>0$, make an obtuse angle with the vector $\vec{b}=-\lambda^2\hat{i}+4\sqrt{2}\hat{j}+4\sqrt{2}\hat{k}$ and an angle $\theta, \frac{\pi}{6}<\theta<\frac{\pi}{2}$, with the positive z-axis. If the set of all possible values of λ is $(\alpha,\beta)-\{\gamma\}$, then $\alpha+\beta+\gamma$ is equal to ____。

Q8. JEE Main 2026 (22 January Shift 1)

Let $\overrightarrow{AB} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\overrightarrow{AD} = \hat{i} + 2\hat{j} + \lambda\hat{k}, \lambda \in \mathbb{R}$. Let the projection of the vector $\vec{v} = \hat{i} + \hat{j} + \hat{k}$ on the diagonal $\overrightarrow{AC}$ of the parallelogram ABCD be of length one unit. If α, β, where $\alpha > \beta$, be the roots of the equation $\lambda^2 x^2 - 6\lambda x + 5 = 0$, then $2\alpha - \beta$ is equal to

(1) 3 (2) 4 (3) 1 (4) 6

Q9. JEE Main 2026 (23 January Shift 2)

Let $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}, \vec{b} = 2\hat{i} + \hat{j} - \hat{k}, \vec{c} = \lambda\hat{i} + \hat{j} + \hat{k}$ and $\vec{v} = \vec{a} \times \vec{b}$. If $\vec{v} \cdot \vec{c} = 11$ and the length of the projection of $\vec{b}$ on $\vec{c}$ is p, then $9p^2$ is equal to

(1) 12 (2) 6 (3) 4 (4) 9

Q10. JEE Main 2026 (23 January Shift 2)

Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $\vec{a} \times \vec{b} = 2(\vec{a} \times \vec{c})$. If $|\vec{a}| = 1, |\vec{b}| = 4, |\vec{c}| = 2$, and the angle between $\vec{b}$ and $\vec{c}$ is 60°, then $|\vec{a} \cdot \vec{c}|$ is equal to

(1) 1 (2) 2 (3) 0 (4) 4

Q11. JEE Main 2026 (21 January Shift 1)

Let $\vec{a} = -\hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b} = 8\hat{i} + 7\hat{j} - 3\hat{k}$ and $\vec{c}$ be vector such that $\vec{a} \times \vec{c} = \vec{b}$. If $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$, then $|\vec{a} + \vec{c}|^2$ is equal to :

(1) 27 (2) 33 (3) 35 (4) 30

Q12. JEE Main 2026 (28 January Shift 1)

For three unit vectors $\vec{a}, \vec{b}, \vec{c}$ satisfying $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$ and $|2\vec{a} + k\vec{b} + k\vec{c}| = 3$, the positive value of k is

(1) 4 (2) 3 (3) 5 (4) 6

Q13. JEE Main 2026 (28 January Shift 1)

Let PQR be a triangle such that $\overrightarrow{PQ} = -2\hat{i} - \hat{j} + 2\hat{k}$ and $\overrightarrow{PR} = a\hat{i} + b\hat{j} - 4\hat{k}, a, b \in \mathbb{Z}$. Let S be the point on QR, which is equidistant from the lines PQ and PR. If $|\overrightarrow{PR}| = 9$ and $\overrightarrow{PS} = \hat{i} - 7\hat{j} + 2\hat{k}$, then the value of $3a - 4b$ is ____

Q14. JEE Main 2026 (24 January Shift 1)

Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}, \vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \vec{a} \times \vec{b}$. Let $\vec{d}$ be a vector such that $|\vec{d} - \vec{a}| = \sqrt{11}, |\vec{c} \times \vec{d}| = 3$ and the angle between $\vec{c}$ and $\vec{d}$ is $\frac{\pi}{4}$. Then $\vec{a} \cdot \vec{d}$ is equal to

(1) 3 (2) 11 (3) 0 (4) 1

ANSWERS AND SOLUTIONS

1. (3)	**2.** (4)	**3.** (2)	**4.** (2)	**5.** (2)	**6.** (3)	**7.** 5	**8.** (1)
9. (1)	**10.** (1)	**11.** (1)	**12.** (3)	**13.** 37	**14.** (3)		

1. **(3)** Given $|\vec{p}| = 2\sqrt{3}$, $|\vec{q}| = 2$, and $\cos\theta = \frac{1}{\sqrt{3}}$ where θ is the angle between $\vec{p}$ and $\vec{q}$.

Since $\vec{p} = \overrightarrow{BC}$, $\vec{q} = \overrightarrow{CA}$, and $\vec{r} = \overrightarrow{BA}$, from triangle closure $\vec{r} = \vec{p} + \vec{q}$.

$\vec{p} \cdot \vec{q} = |\vec{p}||\vec{q}|\cos\theta = 2\sqrt{3} \cdot 2 \cdot \frac{1}{\sqrt{3}} = 4$

$|\vec{r}|^2 = |\vec{p}|^2 + |\vec{q}|^2 + 2\vec{p} \cdot \vec{q} = 12 + 4 + 8 = 24$

$\vec{q} - 3\vec{r} = \vec{q} - 3(\vec{p} + \vec{q}) = -3\vec{p} - 2\vec{q}$

$\vec{p} \times (\vec{q} - 3\vec{r}) = -2(\vec{p} \times \vec{q})$

$\sin^2\theta = 1 - \frac{1}{3} = \frac{2}{3}$

$|\vec{p} \times \vec{q}|^2 = 12 \cdot 4 \cdot \frac{2}{3} = 32$

$|\vec{p} \times (\vec{q} - 3\vec{r})|^2 = 4 \cdot 32 = 128$

Result: $128 + 3(24) = 128 + 72 = 200$

2. **(4)** $\vec{c} = \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 2 \\ 1 & -1 & -3 \end{vmatrix} = (-1, -1, 0)$.

$\vec{d} = \vec{c} \times \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -1 & 0 \\ -1 & 1 & 2 \end{vmatrix} = (-2, 2, -2)$.

$\vec{a} - \vec{b} = (-2, 2, 5)$.

$(\vec{a} - \vec{b}) \cdot \vec{d} = (-2)(-2) + (2)(2) + (5)(-2) = 4 + 4 - 10 = -2$.

3. **(2)** Since P is equidistant from lines AB and AC through point A, it lies along the angle bisector direction. The unit vectors are $\hat{u}_{AB} = \frac{3\vec{i}+\vec{j}-\vec{k}}{\sqrt{11}}$ and $\hat{u}_{AC} = \frac{\vec{i}-\vec{j}+3\vec{k}}{\sqrt{11}}$.

The bisector direction is proportional to $2\vec{i} + \vec{k}$, so $\vec{AP} = \lambda(2\vec{i} + \vec{k})$.

From $|\vec{AP}| = \frac{\sqrt{5}}{2}$: $|\lambda|\sqrt{5} = \frac{\sqrt{5}}{2}$, giving $\lambda = \frac{1}{2}$.

Thus $\vec{AP} = \vec{i} + \frac{1}{2}\vec{k}$.

Area of triangle ABP = $\frac{1}{2}|\vec{AB} \times \vec{AP}|$.

Computing: $\vec{AB} \times \vec{AP} = \frac{1}{2}\vec{i} - \frac{5}{2}\vec{j} - \vec{k}$.

$|\vec{AB} \times \vec{AP}| = \sqrt{\frac{1}{4} + \frac{25}{4} + 1} = \sqrt{\frac{30}{4}} = \frac{\sqrt{30}}{2}$.

Area = $\frac{\sqrt{30}}{4}$.

4. **(2)** $2(\vec{a} \times \vec{c}) + 3(\vec{b} \times \vec{c}) = \vec{0} \Rightarrow (2\vec{a} + 3\vec{b}) \times \vec{c} = \vec{0}$.

So $\vec{c} \parallel (2\vec{a} + 3\vec{b}) = (7, -13, 19)$. Let $\vec{c} = t(7, -13, 19)$.

$\vec{a} - \vec{b} = (1, -4, 2)$.

$(\vec{a}-\vec{b})\cdot\vec{c}=t(7+52+38)=97t=-97\Rightarrow t=-1.$

$\vec{c}=(-7,13,-19).$

$\vec{c}\times\hat{k}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -7 & 13 & -19\\ 0 & 0 & 1\end{vmatrix}=(13,7,0).$

$|\vec{c}\times\hat{k}|^2=169+49=218.$

5. **(2)** $\vec{v}=\lambda\vec{a}+\mu\vec{b}=(2\lambda+\mu,\ -\lambda+3\mu,\ -\lambda-\mu).$

$\vec{v}\cdot\vec{c}=2(2\lambda+\mu)+(-\lambda+3\mu)+3(-\lambda-\mu)=2\mu.$

$|\vec{c}|=\sqrt{4+1+9}=\sqrt{14}.$

Projection length $=\frac{|2\mu|}{\sqrt{14}}=\frac{1}{\sqrt{14}}\Rightarrow|\mu|=\frac{1}{2}.$

$|\vec{v}|^2=6\lambda^2+11\mu^2=6(1)+\frac{11}{4}=\frac{35}{4}.$

$|\vec{v}|=\frac{\sqrt{35}}{2}.$

6. **(3)** $\vec{a}=2\hat{i}-\hat{j}+\hat{k}$

$\vec{b}=\lambda\hat{j}+2\hat{k}\ ;\ \lambda\in Z$

$\vec{c}=\vec{a}\times\vec{b}=(-2-\lambda)\hat{i}-4\hat{j}+2\lambda\hat{k}$

$|\vec{c}|=\sqrt{53}$

$\Rightarrow 5\lambda^2+4\lambda-33=0$

$\lambda=2.2$ or -3

$\Rightarrow\boxed{\lambda=-3}$

$\vec{c}=\hat{i}-4\hat{j}-6\hat{k}$

let $\vec{d}=y\hat{j}+z\hat{k}$

$|\vec{d}|=2$

$\Rightarrow y^2+z^2=4$

$(\vec{c}\cdot\vec{d})^2=(4y+6z)^2\le\left(\sqrt{4^2+6^2}\times\sqrt{y^2+z^2}\right)^2\le 208$

7. **(5)** $\frac{\vec{a}.\hat{k}}{|\vec{a}|}=\cos\theta\Rightarrow\frac{\lambda}{\sqrt{3+\lambda^2}}=\cos\theta$

$\Rightarrow 0<\frac{\lambda}{\sqrt{3+\lambda^2}}<\frac{\sqrt{3}}{2}$

$\Rightarrow\lambda>0\&4\lambda^2<9+3\lambda^2\Rightarrow\lambda^2<9$

$\Rightarrow\lambda\in(0,3)\qquad\ldots.(1)$

$\Rightarrow\vec{a}\cdot\vec{b}<0\Rightarrow-\sqrt{2}\lambda^2-4\sqrt{2}+4\sqrt{2}\lambda<0$

$\Rightarrow\lambda^2-4\lambda+4>0\Rightarrow(\lambda-2)^2>0$

$\Rightarrow\lambda\ne 2$

from $(1)\&(2)$

$\lambda\in(0,3)-\{2\}$

$\therefore\alpha=0,\beta=3,\gamma=2$

$\Rightarrow\alpha+\beta+\gamma=5$

8. **(1)** $\vec{AC} = \vec{AB} + \vec{AD} = 3\hat{i} + 6\hat{j} + (\lambda - 5)\hat{k}$.

Projection of $\vec{v} = \hat{i} + \hat{j} + \hat{k}$ on $\vec{AC}$: $\frac{|\vec{v}\cdot\vec{AC}|}{|\vec{AC}|} = 1$.

$\vec{v} \cdot \vec{AC} = 3 + 6 + (\lambda - 5) = \lambda + 4$, $|\vec{AC}| = \sqrt{45 + (\lambda - 5)^2}$.

$(\lambda + 4)^2 = 45 + (\lambda - 5)^2$

$\Rightarrow 8\lambda + 16 = 45 - 10\lambda + 25$

$\Rightarrow 18\lambda = 54 \Rightarrow \lambda = 3$.

Equation: $9x^2 - 18x + 5 = 0$

$\Rightarrow (3x - 1)(3x - 5) = 0$

$\Rightarrow x = \frac{1}{3}, \frac{5}{3}$.

$\alpha = \frac{5}{3}$, $\beta = \frac{1}{3}$.

$2\alpha - \beta = \frac{10}{3} - \frac{1}{3} = 3$.

9.

(1) $\vec{v} = \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 2 & 1 & -1 \end{vmatrix} = -\hat{i} + 7\hat{j} + 5\hat{k}$.

$\vec{v} \cdot \vec{c} = -\lambda + 7 + 5 = 11 \Rightarrow \lambda = 1$.

$\vec{c} = \hat{i} + \hat{j} + \hat{k}$, $|\vec{c}| = \sqrt{3}$.

$\vec{b} \cdot \vec{c} = 2 + 1 - 1 = 2$.

Projection $p = \frac{|\vec{b}\cdot\vec{c}|}{|\vec{c}|} = \frac{2}{\sqrt{3}}$.

$9p^2 = 9 \times \frac{4}{3} = 12$.

10. **(1)** $\vec{a} \times \vec{b} = 2(\vec{a} \times \vec{c}) \Rightarrow \vec{a} \times (\vec{b} - 2\vec{c}) = \vec{0}$, so $\vec{b} - 2\vec{c} = k\vec{a}$.

$\vec{b} \cdot \vec{c} = 4 \cdot 2 \cdot \cos 60^\circ = 4$.

Dot with $\vec{c}$: $4 = 2(4) + k(\vec{a} \cdot \vec{c}) \Rightarrow k(\vec{a} \cdot \vec{c}) = -4$.

Dot with $\vec{a}$: $\vec{a} \cdot \vec{b} = 2(\vec{a} \cdot \vec{c}) + k$.

Dot with $\vec{b}$: $16 = 8 + k\vec{a} \cdot \vec{b} = 8 + k\left(-\frac{8}{k} + k\right) = k^2 \Rightarrow k = \pm 4$.

$\vec{a} \cdot \vec{c} = -\frac{4}{k} = \mp 1 \Rightarrow |\vec{a} \cdot \vec{c}| = 1$.

11. **(1)** Let $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$.

From $\vec{a} \times \vec{c} = \vec{b}$: $z - y = 4$, $z + 2x = 7$, $y + 2x = 3$.

From $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$: $x + y + z = 4$.

Solving: $y = -1$, $z = 3$, $x = 2$.

$\vec{c} = 2\hat{i} - \hat{j} + 3\hat{k}$.

$\vec{a} + \vec{c} = (-1 + 2)\hat{i} + (2 - 1)\hat{j} + (2 + 3)\hat{k} = \hat{i} + \hat{j} + 5\hat{k}$.

$|\vec{a} + \vec{c}|^2 = 1 + 1 + 25 = 27$.

12. **(3)** Expanding $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$ with unit vectors gives $6 - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 9$.

So $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -3/2$.

For the symmetric case where vectors are equally spaced (e.g., on a circle at 120° intervals), we have

$\vec{a}\cdot\vec{b}=\vec{b}\cdot\vec{c}=\vec{c}\cdot\vec{a}=-1/2$.

Expanding $|2\vec{a}+k\vec{b}+k\vec{c}|^2=9$ gives $4+2k^2+4k(\vec{a}\cdot\vec{b}+\vec{a}\cdot\vec{c})+2k^2\vec{b}\cdot\vec{c}=9$.

Substituting the symmetric values: $4+2k^2(1/2)+4k(-1)+2k^2(-1/2)=9$, which simplifies to $k^2-4k-5=0$.

Factoring: $(k-5)(k+1)=0$, giving $k=5$ as the positive solution.

13. **(37)** Given: $\vec{PQ}=-2\hat{i}-\hat{j}+2\hat{k}$, $\vec{PR}=a\hat{i}+b\hat{j}-4\hat{k}$ $(a,b\in\mathbb{Z})$, $\vec{PS}=\hat{i}-7\hat{j}+2\hat{k}$, $|\vec{PR}|=9$

From $|\vec{PR}|=9$: $a^2+b^2+16=81\Rightarrow a^2+b^2=65$ ...(1)

Since S is on QR equidistant from lines PQ and PR, PS is the angle bisector. So:

$\cos\theta=\frac{\vec{PQ}\cdot\vec{PS}}{|\vec{PQ}||\vec{PS}|}=\frac{-2+7+4}{3\cdot3\sqrt{6}}=\frac{9}{9\sqrt{6}}=\frac{1}{\sqrt{6}}$

Similarly: $\frac{1}{\sqrt{6}}=\frac{\vec{PS}\cdot\vec{PR}}{|\vec{PS}||\vec{PR}|}=\frac{a-7b-8}{3\sqrt{6}\cdot9}$

$a-7b=35$ ...(2)

From (1) and (2): $a=7,b=-4$

$3a-4b=21+16=37$ (NTA Answer)

However, $\vec{QS}=3\hat{i}-6\hat{j}$ and $\vec{SR}=6\hat{i}+3\hat{j}-6\hat{k}$ are not parallel, so Q, S, R are not collinear. Also $\cos\theta=\frac{2\sqrt{2}}{\sqrt{3}}>1$, which is impossible. Hence no such triangle exists.

So, this should be a bonus question.

14.

(3) $\vec{c}=\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&1&-2\\1&1&0\end{vmatrix}=2\hat{i}-2\hat{j}+\hat{k}$

$|\vec{c}|=\sqrt{4+4+1}=3$, $|\vec{a}|=\sqrt{4+1+4}=3$

From $|\vec{c}\times\vec{d}|=3$ and $\cos(\angle(\vec{c},\vec{d}))=\frac{1}{\sqrt{2}}$:

$|\vec{c}||\vec{d}|\sin\frac{\pi}{4}=3\rightarrow3|\vec{d}|\cdot\frac{1}{\sqrt{2}}=3\rightarrow|\vec{d}|=\sqrt{2}$

From $|\vec{d}-\vec{a}|^2=11$:

$|\vec{d}|^2-2\vec{a}\cdot\vec{d}+|\vec{a}|^2=11$

$2-2\vec{a}\cdot\vec{d}+9=11$

$\vec{a}\cdot\vec{d}=0$